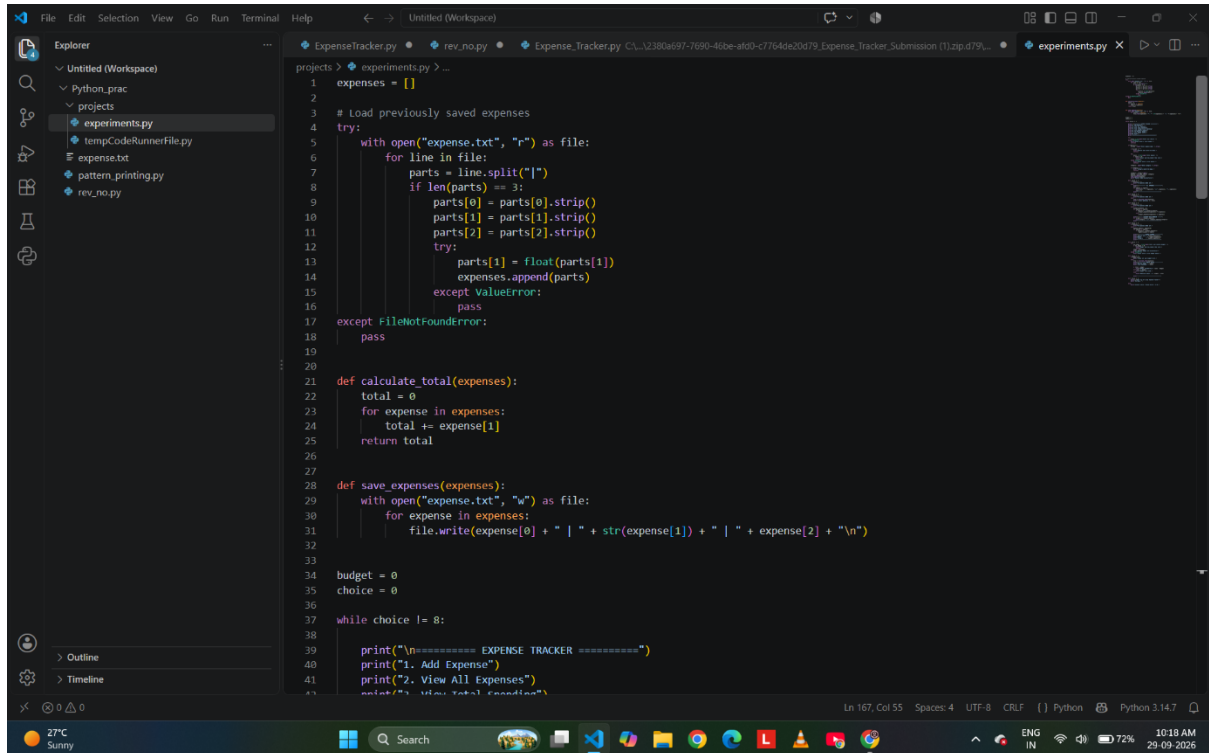


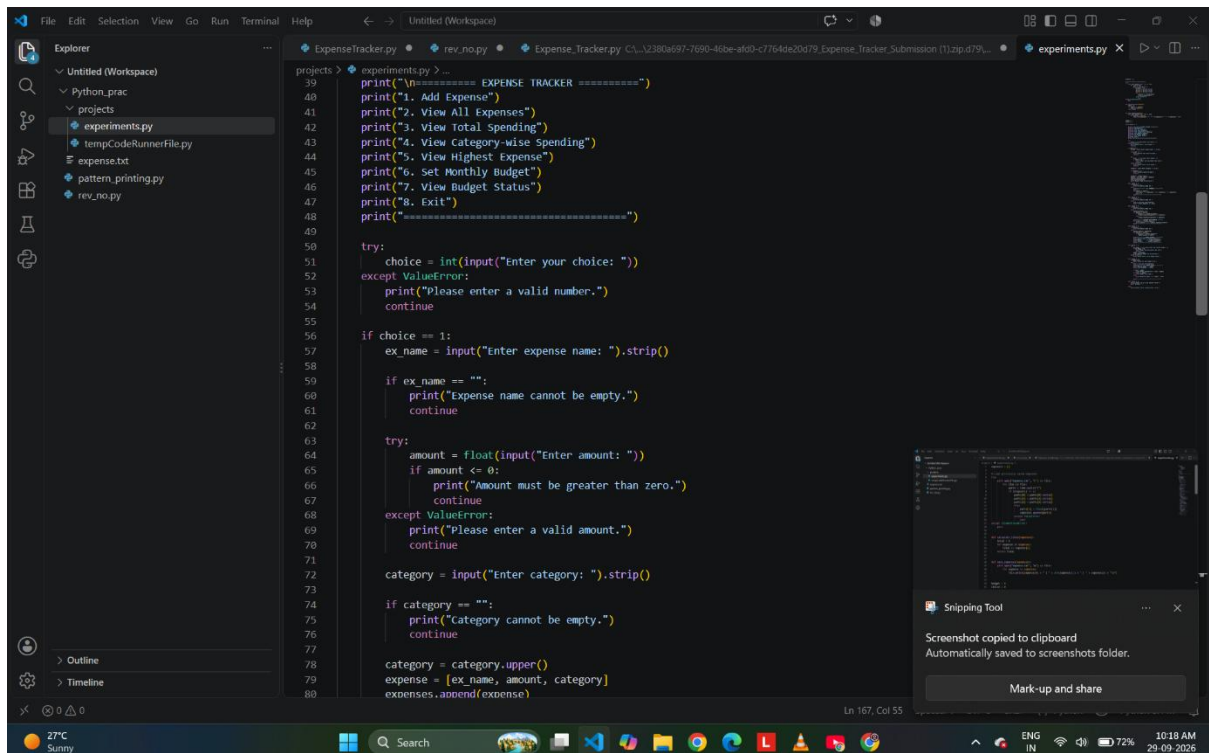
SCREENSHOTS

Some screenshots of the program :-



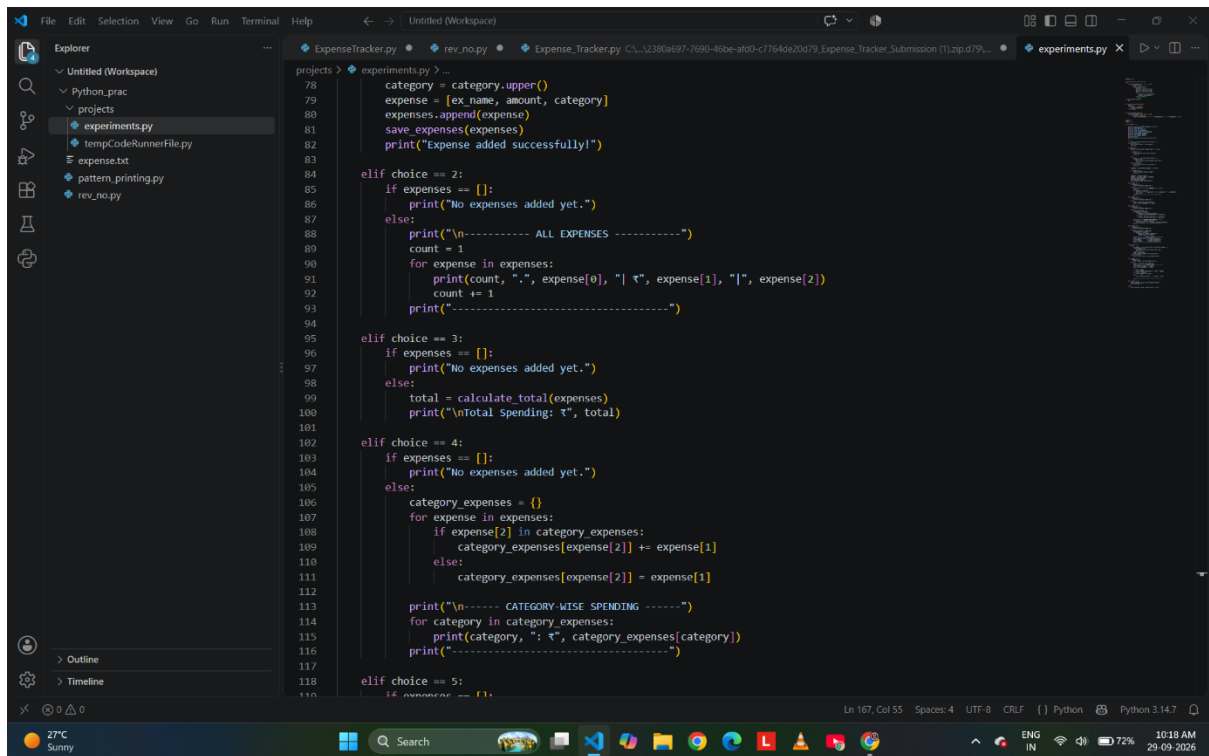
This screenshot shows the initial setup of an expense tracker program in a code editor. The Explorer panel on the left shows a project structure with files like `experiments.py`, `tempCodeRunnerFile.py`, `expense.txt`, `pattern_printing.py`, and `rev_no.py`. The main editor displays the `experiments.py` file, which includes functions for loading expenses from a file, calculating the total, and saving expenses. The program starts with a menu loop where the user can choose to add an expense, view all expenses, view total spending, or exit.

```
1 expenses = []
2
3 # Load previously saved expenses
4 try:
5     with open("expense.txt", "r") as file:
6         for line in file:
7             parts = line.split("|")
8             if len(parts) == 3:
9                 parts[0] = parts[0].strip()
10                parts[1] = parts[1].strip()
11                parts[2] = parts[2].strip()
12                try:
13                    parts[1] = float(parts[1])
14                    expenses.append(parts)
15                except ValueError:
16                    pass
17 except FileNotFoundError:
18     pass
19
20
21 def calculate_total(expenses):
22     total = 0
23     for expense in expenses:
24         total += expense[1]
25     return total
26
27
28 def save_expenses(expenses):
29     with open("expense.txt", "w") as file:
30         for expense in expenses:
31             file.write(expense[0] + " | " + str(expense[1]) + " | " + expense[2] + "\n")
32
33
34 budget = 0
35 choice = 0
36
37 while choice != 8:
38
39     print("\n===== EXPENSE TRACKER =====")
40     print("1. Add Expense")
41     print("2. View All Expenses")
42     print("3. View Total Spending")
43     print("4. View Category-wise Spending")
44     print("5. View Highest Expense")
45     print("6. Set Monthly Budget")
46     print("7. View Budget Status")
47     print("8. Exit")
48     print("=====")
49
50     try:
51         choice = int(input("Enter your choice: "))
52     except ValueError:
53         print("Please enter a valid number.")
54         continue
55
56     if choice == 1:
57         ex_name = input("Enter expense name: ").strip()
58
59         if ex_name == "":
60             print("Expense name cannot be empty.")
61             continue
62
63         try:
64             amount = float(input("Enter amount: "))
65             if amount <= 0:
66                 print("Amount must be greater than zero.")
67                 continue
68         except ValueError:
69             print("Please enter a valid amount.")
70             continue
71
72         category = input("Enter category: ").strip()
73
74         if category == "":
75             print("Category cannot be empty.")
76             continue
77
78         category = category.upper()
79         expense = [ex_name, amount, category]
80         expenses.append(expense)
```

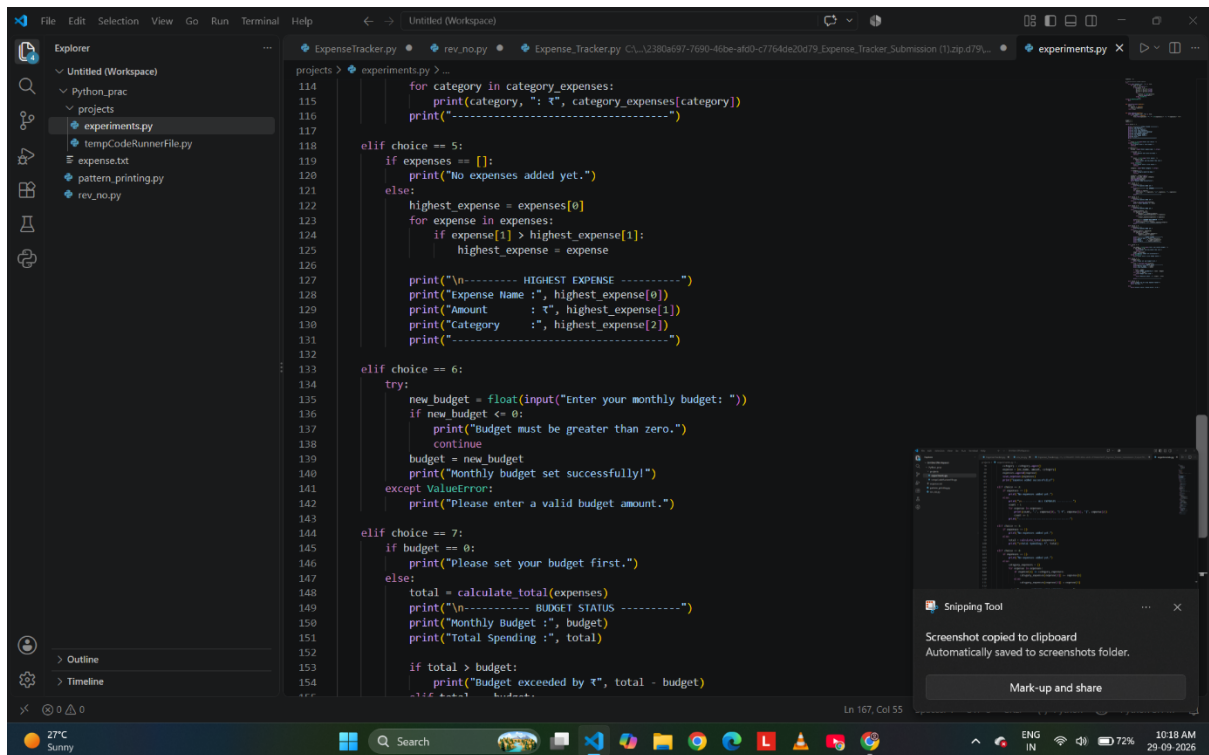


This screenshot shows the continuation of the expense tracker program. The code continues from the previous screenshot, handling the user's choice to add an expense. It includes input validation for the expense name, amount, and category. The program then appends the new expense to the `expenses` list. A Snipping Tool window is visible in the bottom right corner, indicating that a screenshot was taken.

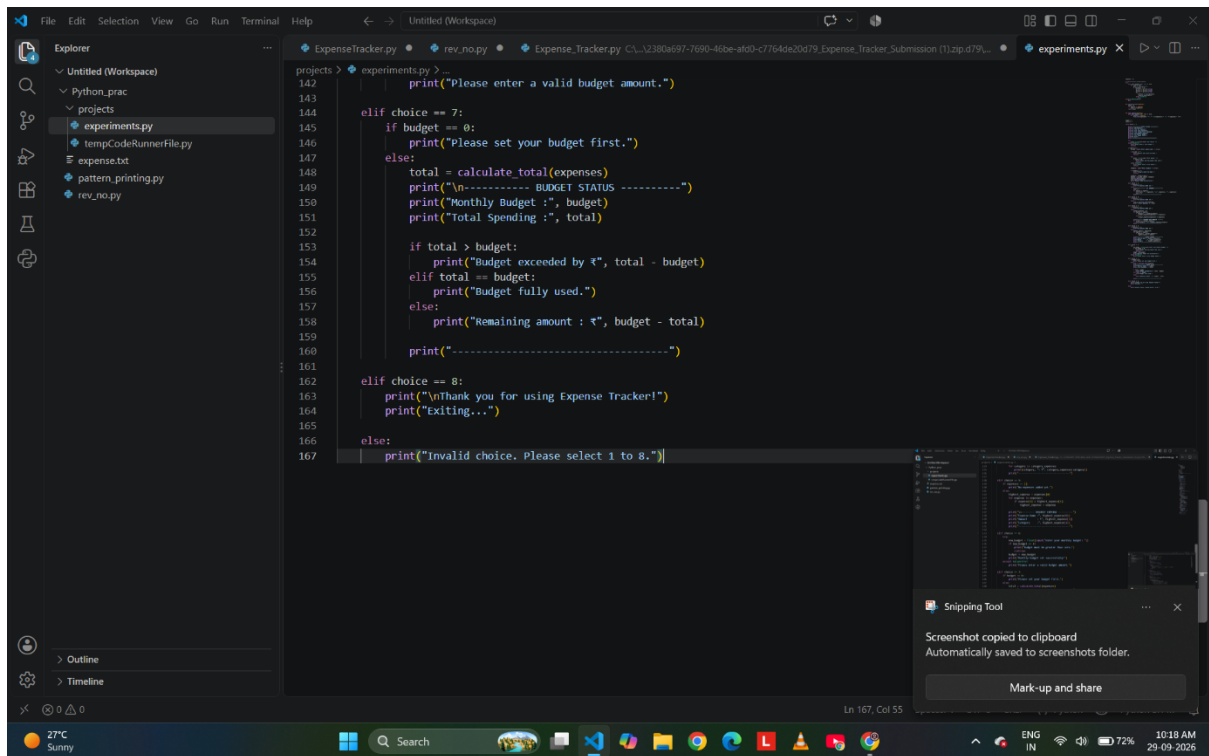
```
39 print("\n===== EXPENSE TRACKER =====")
40 print("1. Add Expense")
41 print("2. View All Expenses")
42 print("3. View Total Spending")
43 print("4. View Category-wise Spending")
44 print("5. View Highest Expense")
45 print("6. Set Monthly Budget")
46 print("7. View Budget Status")
47 print("8. Exit")
48 print("=====")
49
50 try:
51     choice = int(input("Enter your choice: "))
52 except ValueError:
53     print("Please enter a valid number.")
54     continue
55
56 if choice == 1:
57     ex_name = input("Enter expense name: ").strip()
58
59     if ex_name == "":
60         print("Expense name cannot be empty.")
61         continue
62
63     try:
64         amount = float(input("Enter amount: "))
65         if amount <= 0:
66             print("Amount must be greater than zero.")
67             continue
68     except ValueError:
69         print("Please enter a valid amount.")
70         continue
71
72     category = input("Enter category: ").strip()
73
74     if category == "":
75         print("Category cannot be empty.")
76         continue
77
78     category = category.upper()
79     expense = [ex_name, amount, category]
80     expenses.append(expense)
```



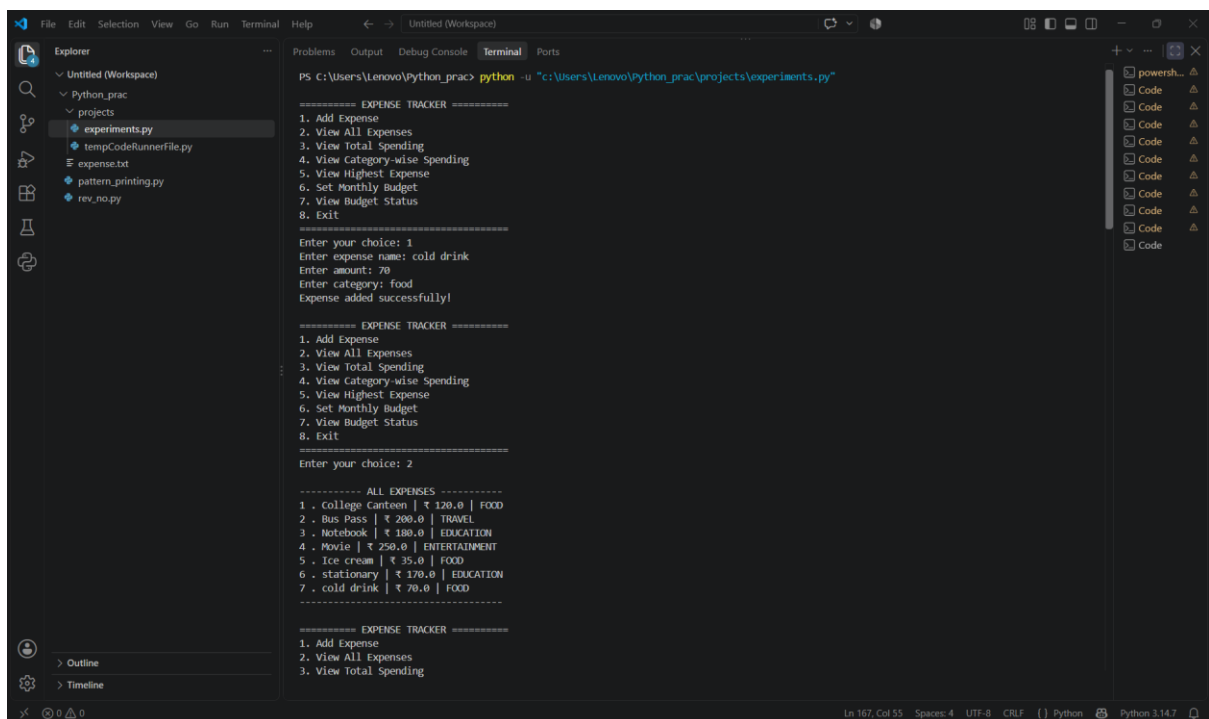
```
78     category = category.upper()
79     expense = [ex_name, amount, category]
80     expenses.append(expense)
81     save_expenses(expenses)
82     print("Expense added successfully!")
83
84 elif choice == 2:
85     if expenses == []:
86         print("No expenses added yet.")
87     else:
88         print("\n----- ALL EXPENSES -----")
89         count = 1
90         for expense in expenses:
91             print(count, ".", expense[0], "| ₹", expense[1], "|", expense[2])
92             count += 1
93         print("-----")
94
95 elif choice == 3:
96     if expenses == []:
97         print("No expenses added yet.")
98     else:
99         total = calculate_total(expenses)
100        print("\nTotal Spending: ₹", total)
101
102 elif choice == 4:
103     if expenses == []:
104         print("No expenses added yet.")
105     else:
106         category_expenses = {}
107         for expense in expenses:
108             if expense[2] in category_expenses:
109                 category_expenses[expense[2]] += expense[1]
110             else:
111                 category_expenses[expense[2]] = expense[1]
112
113         print("\n----- CATEGORY-WISE SPENDING -----")
114         for category in category_expenses:
115             print(category, ": ₹", category_expenses[category])
116         print("-----")
117
118 elif choice == 5:
119     if expenses == []:
120         print("No expenses added yet.")
121     else:
122         highest_expense = expenses[0]
123         for expense in expenses:
124             if expense[1] > highest_expense[1]:
125                 highest_expense = expense
126
127         print("\n----- HIGHEST EXPENSE -----")
128         print("Expense Name :", highest_expense[0])
129         print("Amount      : ₹", highest_expense[1])
130         print("Category    :", highest_expense[2])
131         print("-----")
132
133 elif choice == 6:
134     try:
135         new_budget = float(input("Enter your monthly budget: "))
136         if new_budget <= 0:
137             print("Budget must be greater than zero.")
138             continue
139         budget = new_budget
140         print("Monthly budget set successfully!")
141     except ValueError:
142         print("Please enter a valid budget amount.")
143
144 elif choice == 7:
145     if budget == 0:
146         print("Please set your budget first.")
147     else:
148         total = calculate_total(expenses)
149         print("\n----- BUDGET STATUS -----")
150         print("Monthly Budget is, budget")
151         print("Total Spending is, total")
152
153         if total > budget:
154             print("Budget exceeded by ₹", total - budget)
```



```
114         for category in category_expenses:
115             print(category, ": ₹", category_expenses[category])
116         print("-----")
117
118 elif choice == 5:
119     if expenses == []:
120         print("No expenses added yet.")
121     else:
122         highest_expense = expenses[0]
123         for expense in expenses:
124             if expense[1] > highest_expense[1]:
125                 highest_expense = expense
126
127         print("\n----- HIGHEST EXPENSE -----")
128         print("Expense Name :", highest_expense[0])
129         print("Amount      : ₹", highest_expense[1])
130         print("Category    :", highest_expense[2])
131         print("-----")
132
133 elif choice == 6:
134     try:
135         new_budget = float(input("Enter your monthly budget: "))
136         if new_budget <= 0:
137             print("Budget must be greater than zero.")
138             continue
139         budget = new_budget
140         print("Monthly budget set successfully!")
141     except ValueError:
142         print("Please enter a valid budget amount.")
143
144 elif choice == 7:
145     if budget == 0:
146         print("Please set your budget first.")
147     else:
148         total = calculate_total(expenses)
149         print("\n----- BUDGET STATUS -----")
150         print("Monthly Budget is, budget")
151         print("Total Spending is, total")
152
153         if total > budget:
154             print("Budget exceeded by ₹", total - budget)
```



Screenshots of Output :-



```
File Edit Selection View Go Run Terminal Help
Untitled (Workspace)

Explorer
  Untitled (Workspace)
  Python_prac
    projects
      experiments.py
      tempCodeRunnerFile.py
      expense.txt
      pattern_printing.py
      rev_no.py

Problems Output Debug Console Terminal Ports
===== EXPENSE TRACKER =====
1. Add Expense
2. View All Expenses
3. View Total Spending
4. View Category-wise Spending
5. View Highest Expense
6. Set Monthly Budget
7. View Budget Status
8. Exit

Enter your choice: 3

Total Spending: ₹ 1025.0

===== EXPENSE TRACKER =====
1. Add Expense
2. View All Expenses
3. View Total Spending
4. View Category-wise Spending
5. View Highest Expense
6. Set Monthly Budget
7. View Budget Status
8. Exit

Enter your choice: 4

----- CATEGORY-WISE SPENDING -----
FOOD : ₹ 225.0
TRAVEL : ₹ 200.0
EDUCATION : ₹ 350.0
ENTERTAINMENT : ₹ 250.0

===== EXPENSE TRACKER =====
1. Add Expense
2. View All Expenses
3. View Total Spending
4. View Category-wise Spending
5. View Highest Expense
6. Set Monthly Budget
7. View Budget Status
8. Exit

Ln 167, Col 55  Spaces: 4  UTF-8  CRLF  Python  Python 3.14.7
```

```
File Edit Selection View Go Run Terminal Help
Untitled (Workspace)

Explorer
  Untitled (Workspace)
  Python_prac
    projects
      experiments.py
      tempCodeRunnerFile.py
      expense.txt
      pattern_printing.py
      rev_no.py

Problems Output Debug Console Terminal Ports

===== EXPENSE TRACKER =====
1. Add Expense
2. View All Expenses
3. View Total Spending
4. View Category-wise Spending
5. View Highest Expense
6. Set Monthly Budget
7. View Budget Status
8. Exit

Enter your choice: 5

----- HIGHEST EXPENSE -----
Expense Name : Movie
Amount      : ₹ 250.0
Category    : ENTERTAINMENT

===== EXPENSE TRACKER =====
1. Add Expense
2. View All Expenses
3. View Total Spending
4. View Category-wise Spending
5. View Highest Expense
6. Set Monthly Budget
7. View Budget Status
8. Exit

Enter your choice: 6
Enter your monthly budget: 3000
Monthly budget set successfully!

===== EXPENSE TRACKER =====
1. Add Expense
2. View All Expenses
3. View Total Spending
4. View Category-wise Spending
5. View Highest Expense
6. Set Monthly Budget
7. View Budget Status
8. Exit

Ln 167, Col 55  Spaces: 4  UTF-8  CRLF  Python  Python 3.14.7
```

